

Shahid Khan

New Paltz, NY | 315-746-1247 | shahidhkhan.dev@gmail.com
Linkedin: [shahid-h-khan](#) | GitHub: [ShahidHKhan](#) | Website: [shahidhkhan.vercel.app](#)

EDUCATION

State University of New York, New Paltz

Master of Science in Computer Science

New Paltz, NY
Jan. 2027 – May 2028 (Expected)

State University of New York, New Paltz

Bachelor of Science in Computer Science – Cumulative GPA: 3.5/4.0

New Paltz, NY
Aug. 2023 – Dec. 2026

Certifications: Google Cloud Generative AI Leader (2025), AI Engineer Core Track (2026)

Awards: 2026 SUNY New Paltz Hackathon – 1st Place

Research: Architecture of Modern LLMs (2026)

PROFESSIONAL EXPERIENCE

Information Technology Technician – SUNY New Paltz

Aug. 2025 – Present

- Resolved ~30 tickets/week for 100+ users by debugging hardware, software, and OS issues in **TeamDynamix**.
- Secured campus user access by managing accounts, permissions, and password resets in **Microsoft Azure**.
- Prepared new laptops and devices for day-one use by imaging, configuring, and deploying them to faculty and staff.
- Restored Wi-Fi, printer, and classroom AV service by diagnosing connectivity and hardware issues on-site.

AI Engineer — IT Desk – SUNY New Paltz

Jun. 2026 – Aug. 2026

- Cut resolution time on difficult tickets by 30% by designing and deploying “HawkEye,” an LLM-powered RAG backend service.
- Halved schedule generation time (4 weeks to 2) by building “Desk-Scheduler,” an LLM and constraint-solver pipeline.
- Improved HawkEye’s retrieval quality by 25% by benchmarking two embedding designs and shipping the stronger one.
- Drove adoption in a 20-person IT office (2–3 uses/day) by monitoring usage, applying feedback, and cutting latency/cost.

AI Quality Engineer Intern – GridDevs

Sept. 2025 – Dec. 2025

- Identified the most reliable retrieval configuration for a RAG system, as measured by **MRR**, **nDCG**, and **keyword coverage**, by benchmarking multiple embedding models and LLMs against each other.
- Assessed LangChain-based versus native code implementations of the RAG pipeline, evaluating each against performance, stability, and maintainability criteria to inform quality decisions.
- Built automated Gradle test suites to validate data integrations and catch regressions across pipeline iterations.

PROJECTS

HawkEye – Python, LangChain, Gradio, PostgreSQL/pgvector, Supabase

github.com/ShahidHKhan/HawkEye

- Improved retrieval accuracy (**MRR**, **nDCG**) on a 100-question eval set over an 850+ article knowledge base via query expansion, source merging, and LLM reranking.
- Migrated the vector store from ChromaDB to Postgres/pgvector on Supabase, writing a backfill script to safely re-embed 4,000+ chunks with duplicate-run protection.
- Built a weekly **GitHub Actions** CI/CD pipeline that re-embeds only new or changed articles, with a circuit breaker that aborts suspicious mass deletions.

Desk-Scheduler – Python, LangGraph, Gemini, OR-Tools, Streamlit, SQLite, Docker

github.com/ShahidHKhan/scheduler

- Architected a containerized **LangGraph** workflow using the **Gemini API** to extract structured JSON availability data from unstructured PDFs, spreadsheets, and photos.
- Formulated a deterministic **OR-Tools (CP-SAT)** constraint model, separating LLM parsing from provable assignment logic to prevent hallucinated schedules.
- Verified all 6 hard scheduling rules with a 16-test pytest suite and diagnosed infeasibility via an **LLM-as-a-judge**.

AutoPricer – Python, PyTorch, XGBoost, FastAPI, Modal

github.com/ShahidHKhan/AutoPricer

- Cut baseline MAE 71% (\$9,781 to \$2,799) on 500K used-car listings with tuned XGBoost; a multi-agent ensemble (QLoRA Llama-3.2-3B + RAG Gemini over 300K embeddings) cut it a further 25% (\$1,599 to \$1,202).
- Cut Llama-3.2-3B’s pricing error by 63% (MAE \$6,723 to \$2,471) via QLoRA fine-tuning on a free Colab T4 GPU (4-bit NF4, 18M-parameter adapter).
- Deployed inference as a streaming FastAPI service on Modal (T4 GPU), integrated with a Chrome extension that flags mispriced cars from live marketplace data.

SKILLS

LLM Engineering: RAG, Reranking, LangChain, LangGraph, Agents/Tool Calling, Pydantic, LLM-as-a-Judge Evals

Machine Learning: PyTorch, Hugging Face (TRL, QLoRA), scikit-learn, XGBoost, Weights & Biases, Google OR-Tools

Backend & Frameworks: Python, TypeScript/JavaScript, FastAPI, Node.js, Streamlit, Gradio, REST APIs, JSON

Databases & Deployment: SQL, PostgreSQL (pgvector), Supabase, SQLite, ChromaDB, Modal, Docker, Git, Azure